

Developing an AI-powered System for Meta-software Development

MARKS: This component is worth 10% of the coursework.

DEADLINE: Between May 18th and 29th, 2026. More details about the timeslot booking will be provided during the lab session and the module announcement on May 15th 2026.

WHERE: In-person colloquium event.

NOTE: Your delivery is in person; however, you should also submit your presentation electronically as a PDF to Learning Mall. Your PDF file should be named with your student ID and name in the format **StudentID-Your_Name.zip**. For example, **12345-Netza_Cruz.zip**

TASK: You should prepare a presentation using one aid to **critically reflect on different software development methodologies**, drawing on insights **from the methodology you have chosen to develop your CW-Software-Component**. The presentation should last approximately 5 minutes, comprising 3 minutes for the presentation and 2 minutes to answer 2 questions.

Your presentation will be assessed during the assessment period. Peers are expected to attend.

Your presentation should demonstrate your ability to:

- **Present and defend** your reflective aid to peers in an authentic professional context.
- **Critically reflect on and evidence** your ability to select and apply appropriate software development methodologies, demonstrating informed decision-making and alignment with system requirements and constraints.
- **Critically reflect on your approach** to the design and modelling of software systems, evidencing the effective use of UML and AI-specific tools to support clarity, structure, and communication of system architectures.
- **Critically reflect on how the integration** of AI-specific tools influences software design choices, modelling practices, and methodological selection, including benefits and limitations.

Learning Outcome	Description
B	Demonstrate the ability to select and apply appropriate software development methodologies
C	Design and Model Software Systems using UML and AI-Specific Tools

Presentation Requirements

1. You have 5 minutes in total for your session (presentation plus Q&A)
 - a. Your presentation should be no more than 3 minutes.
 - b. Your Q&A section should be approximately 2 minutes.
2. Your reflective aid may be a poster, digital poster, or written text. If you have an alternative reflective aid you would like to create, please talk to the module team.
 - a. You **may not** use notes or cue cards to assist you in your presentation.
3. You must submit your presentation electronically in PDF format.

ADVICE

To succeed at conducting your presentation, consider an appropriate weighting of the topics, content and time in relation to the learning outcomes being assessed.

In the **3-minute presentation section**, you are assessed on the quality of reflection in your reflective aid and your verbal reflection in relation to the two learning outcomes.

In the **2-minute Q&A section**, you are assessed on the quality of your answers.

How you use the 3-minute presentation time is up to you, but you should make sure to include **both** a reflective aid and verbal reflection.

REFERENCING SOURCES

You must reference any sources you use. This means that you must include in-text citations and include a complete reference list at the end of your work. For further guidance on the APA Referencing System, please refer to the [XJTLU Library guide](#). Note that referencing should be consistent, accurate, and adhere to XJTLU's Academic Integrity Policy.

ACADEMIC INTEGRITY

Your assignment must be all your own work. Plagiarism, copying, collusion, or dishonest use of data will be penalised. Penalties ranging from capped scores to an award of 0 (zero) will apply. Please refer to the for guidance.

Be advised that more serious instances of academic infringement will be presented at exam boards, which can result in HoDs being notified.

GENERATIVE AI

The written content should be your own work, and GenAI should not be used to create the written content for your visual aid or to prepare a script for your presentation.

STUDY SKILLS

For study skills support, please see the University of Liverpool's [KnowHow](#) service. The section on [reflective writing](#) is useful for all assignments, and the [presentation](#) tutorial is especially relevant for PGC405.

LATENESS

XJTLU policy is -5% per day up to a total of 25%. Work submitted more than five working days late will receive a grade of zero.

Marking Criteria

Criteria	Description	Maximum Marks
Dissemination	Readability and formatting of the aid.	1
	Presentation is structured.	2
	Demonstrates understanding rather than memorisation.	1
Reflection on SDLC	Reflects on the selection and application of appropriate software development methodologies.	1
	Demonstrates understanding rather than memorisation.	1
Reflection on the use of UML	Reflects on the effective use of UML.	1
	Demonstrates understanding rather than memorisation.	1
Reflection on the use of AI tools	Reflects on the integration of AI-specific tools influences software design choices.	1
	Demonstrates understanding rather than memorisation.	1
		10

Appendix 1: Analysis of the Generative AI Usage Policy for the DTS114TC Presentation-Component's Assessment

This policy is designed to be both **permissive** and **protective**. It acknowledges the growing role of AI as a professional tool while safeguarding the module's academic integrity and core learning objectives.

The policy clearly distinguishes between using AI as an **assistant** and using it as a **substitute** for your own intellectual effort.

1. Permitted Uses: AI as a Collaborative Tool

You are encouraged to use Generative AI to *enhance* your work, not create it. The allowed uses align with common tasks where technology can improve efficiency and quality.

- **a) Brainstorming for Inspiration:**
 - **What it means:** You can use AI (e.g., XIPU AI, ChatGPT, DeepSeek, or the module-developed AI agent) to generate initial ideas, mind maps, potential lesson plan structures, or arguments for and against a topic.
 - **Why it's permitted:** This mimics a discussion with a colleague. The final direction, critical selection, and development of ideas must be your own.
- **b) Proof-reading and Language Editing:**
 - **What it means:** You can use AI (e.g., Grammarly, XIPU AI, ChatGPT) specifically to check grammar, spelling, punctuation, and sentence clarity. It can help rephrase awkward sentences *for clarity only*.
- **c) Finding or Generating Images:**
 - **What it means:** You can use AI image generators to create custom illustrations, diagrams, or concept art for your assignment, provided it is relevant.
 - **Why it's permitted:** Visuals can enhance a presentation or report. This is seen as a digital design skill. You must ensure the images are appropriate and correctly referenced if required by the assignment brief.
- **d) Designing Layout:**
 - **What it means:** You can use AI tools to get suggestions on how to structure a document, presentation, or poster for visual appeal and professional presentation.
 - **Why it's permitted:** This focuses on the aesthetics and organisation of information, not the information itself.

2. Prohibited Use: AI as an Author

This is the most critical rule and is non-negotiable.

- **You CANNOT use Generative AI to write the assessments for you.**

- **What it means:** You must not prompt an AI to generate entire paragraphs, sections, or the entire answer to an assessment question and then submit it as your own work. This constitutes academic misconduct (plagiarism).
- **Consequence:** The policy explicitly states that submitted work will be run through AI-writing detection software (e.g., Turnitin AI Detection). If a significant portion of your text is identified as AI-generated, you will likely be investigated for a breach of academic integrity, which can have serious consequences, including failing the module.

Why This Policy Exists

The DTS114TC module aims to develop your **critical thinking, reflective practice, and ability to synthesise theory**.

- **The Process is the Product:** The value of the assessment lies in *your* cognitive process: analysing reading materials, reflecting on your classroom practice, constructing your own arguments, and demonstrating your understanding. Using AI to shortcut this process defeats the entire purpose of the postgraduate certificate.
- **Professional Preparedness:** The policy teaches you to use AI *ethically* and *effectively*, a crucial skill for 21st-century education.
- **Authentic Assessment:** The university needs to verify that *you* have mastered the learning outcomes, not an AI model.

Practical Advice for You as an In-Service Teacher

1. **Always Acknowledge and Reflect:** Even for permitted uses, it is best practice to include a short, honest "**AI Use Statement**" at the end of your assignment.
 - Example: "I used ChatGPT (GPT-4) for brainstorming initial ideas for my pedagogical research topics and for proofreading my final document to correct grammatical errors. All substantive content and critical analysis are my own work."
 - This demonstrates transparency and good academic practice.
2. **You Are the Expert:** Use AI to support and structure your unique insights, not replace them. An AI can give you a generic plan, but it cannot replicate the nuanced understanding you have of your specific context.
3. **Verify Everything:** AI is known for "hallucinating" (making up facts, references, and quotes). Never trust AI-generated information without verifying it against your module's core readings and reputable academic sources.

In summary, this policy encourages you to **use AI as a smart assistant, not a ghostwriter**. Embrace it for enhancing efficiency and presentation, but ensure the intellectual heart of the work, the analysis, reflection, and argument, which is authentically and unmistakably yours.